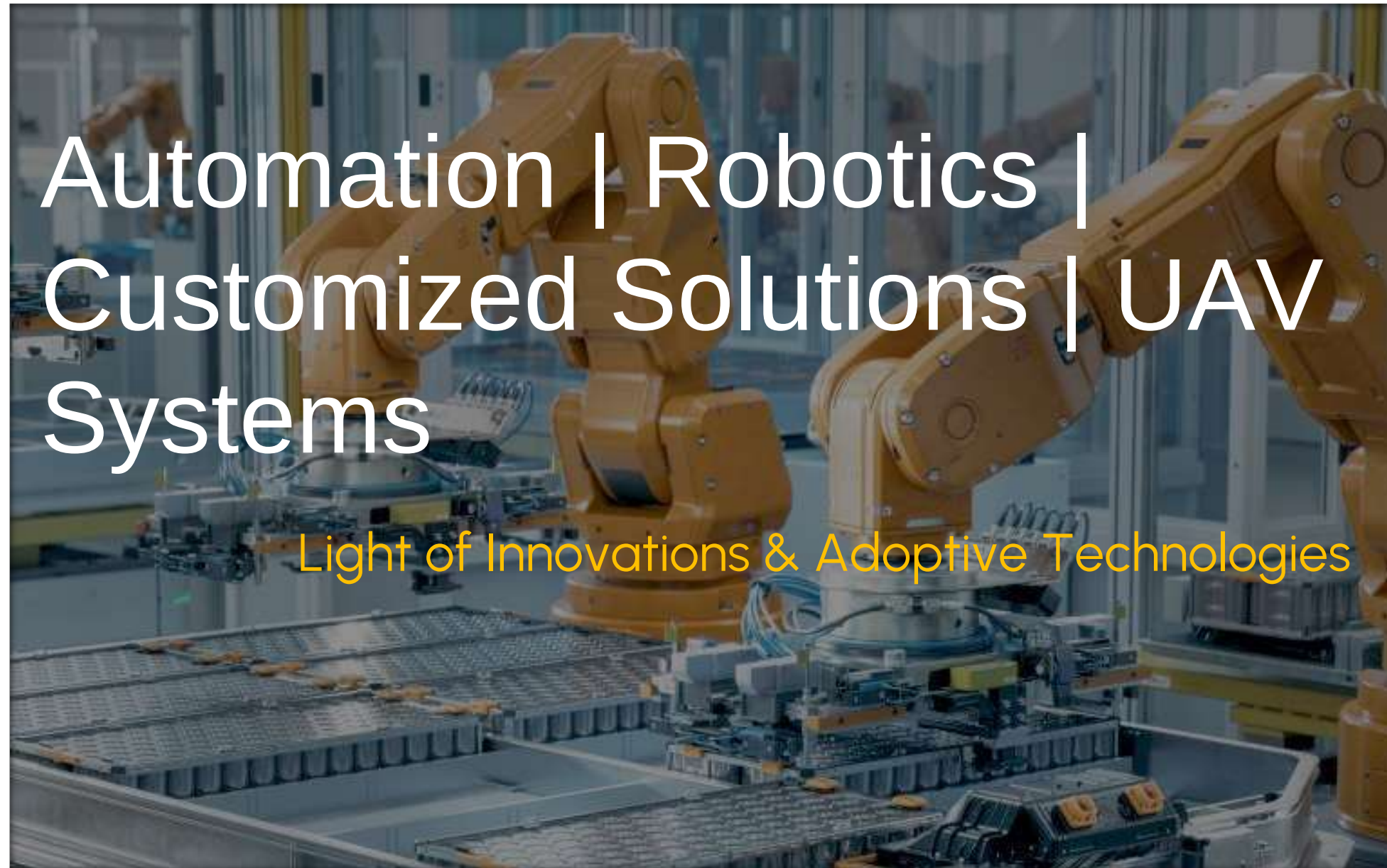
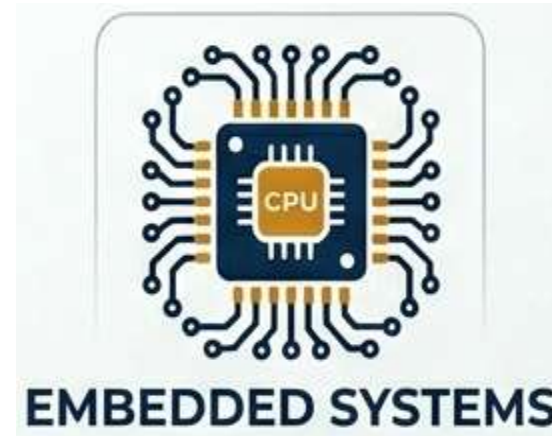




Automation | Robotics | Customized Solutions | UAV Systems

Light of Innovations & Adoptive Technologies

Our Presence



Laser Marking & Cutting Solution



Parameter	Description	Remark
Laser	RAYCUS Photonics fiber optic	
Laser Power	20W	
Mode of Operation	Automatic Pulsating	
Max. Output Power	900W	
Source	Ytterbium fiber	
Power Tunability	10% to 100%	
Pulse Repetition Rate	20KHz to 200KHz	
Depth	Material dependent / 40um for SS316 in single stroke	To achieve desired depth, required multiple passes. There is no any thumb rule. After 2cd pass, it will start removing 20um as the focal length changed.
Operating Temperature	0 to 40 Deg. C	
Lens	HS Optic	
Wave Length	1064	
Pulse duration @20kHz	100ns	
Class (Safety)	IV	
Power Stability	1 to 3%	
Pulse Energy @20kHz	85%/ 80W	
Inbuilt Guide Laser	WL 450nm, Yes, appro 2W	
Beam Quality	>1.6 m2	
Marking Area	100mm*100mm	wave legth & power value exact
Source Life	100000 hrs	
Resolution	100um	
Repeatability	20um	
Beam (Ø)	25 to 30 um	cable dia: ID 5mm fiber, 2 mtr cable length
Cooling System	Active Air cooling	
Power Supply	230VAC, 50Hz	
Min Line Width	25 to 30um	
Hand Held Unit Weight	8 Kg	
Hand Held Unit Dimension	100*150*580mm	without focus stand ; Dia 100*160mm focus stand. Column height is 720 mm



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AI-based Machine Vision System



Sr.No.	Technical Specification
1	Conveyor based Inspection system includes bowl feeder, part locator & separator,
2	Camera for top, bottom & 4 sides of job : 6 cameras
3	2d laser profiler : 4 nos for measurement
4	Job feeding mechanism
5	lumitrax lighting system:
6	Mechanical structure, sensors
7	nvidia GPU, screen & peripheral
8	Control panel as mentioned in SOR
9	Designing architecture, Lib
10	IP creation for Machine learning database
11	User interface, PC application & database
12	ML of 4 components & AI
13	Rejection mechanism



Customized Welding Robot System



Technical Specifications

1. OTG MIG welding robot with 300A power source
2. Inbuilt Data logging system with external sensors like Pressure, Gas flow, Current & Voltage measurement
3. Power Supply AC440V
4. Job holding Fixture with 270° rotating mechanism
5. 10" HMI for smooth operation & parameters settings
6. 5sps Data logging facility & software interface



Automatic Pressure Control System



1	Pressure Range	0-1	psi	As per the capacity of the pressure sensor
2	Pressure set points	0.8	psi	As per the capacity of the pressure sensor
		0.6	psi	As per the capacity of the pressure sensor
		0.4	psi	As per the capacity of the pressure sensor
3	Air Blower motor(Cylindrical)	1.8KW	HP	High Pressure Electric Pump, Model No. HL-8101, Input 220-240VAC, 1800W
4	Motor voltage	220-240	VAC	As mentioned above
5	Phase	Single		210-240VAC
6	Load driving	Contactor	2 O/P REQUIRED	1. Contactor for motor operation; 2. Solenoid valve
7	Input supply available	230VSC		
8	Display	2 line/4 line		3.2" color display
9	Resolution	0.01PSI		0.01 psi
10	Accuracy	1%		± 1 %
11	Operation	Touch		Resistive touch operation
12	Operating temp.	55DEG C		Max 55°C
13	Solenoid Valve	0-2 psi		Make: Aira Euro Automation, Model : SS3; Supply: 230VAC; Pressure: 0-2Kg
14	Enclosure size	150X150X100		Controller fitting box
15	IP rating if any	Water/rain proof		Controller encloser only
16	Additional Accessories			Movable Trolley for controller & blower mounting



Industry 4.0 for Welding machines



1. Selection of hardware & transducers/sensors.
2. Manufacturing of Customized sensor if unavailable.
3. Application/ mounting of sensors on machine.
4. Manufacturing control panels.
5. Programming of Controller
6. Converting Raw data to desire readable form.
7. Programming of HMI.
8. Pushing data on data bus.
9. Validating data on platform
10. Logging & analysis



W 01 SPOT WELD MACHINE				SETTING
W.O. NUMBER	2222		PRESSURE (BAR)	5.8
W.O. QTY.	4000		PRI WELD CURRENT	28.00 A
OPERATOR CODE	7226		FORCE BY PRESSURE	478 KG
NO.OF STROKE	737		NO.OF JOB	4245
WELD TIME (MS)	41		SQUE. TIME (MS)	20
HOLD TIME (MS)	8		FORCE BY LOAD CELL	101 KG
SEC. WELD VOLTAGE	0.00 A	NEXT	SEC. WELD CURRENT	0.00 A
WATER FLOW	2.79 LPM	T	OP. NO.	

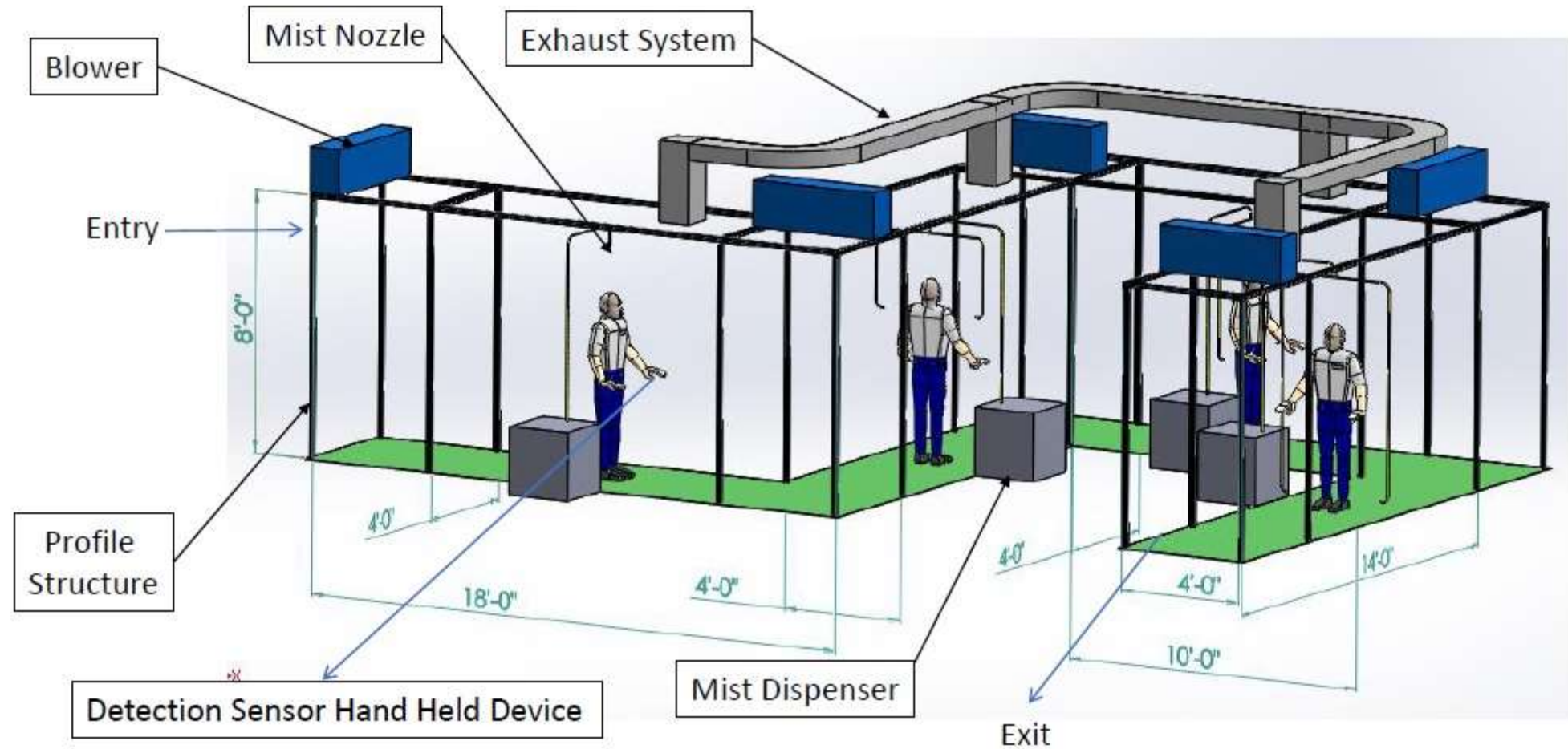


CBRN Military Engineering Chemical Test Tunnel



1. Four Chamber Test Tunnel.
2. 10um chemical droplet spraying system.
3. High pressure chemical pumps.
4. Toxic Chemical detection Sensor .
5. Programming of Controller
6. Spraying after Human detection.
7. Programming of HMI.
8. Pre-programmable system for multiple chemical.
9. Suction facility to clean the tunnel after test
10. Power supply 440VAC

CBRN Military Engineering Chemical Test Tunnel



CBRN Military Engineering Chemical Test Tunnel



Simulation of Chemicals in Closed Chamber, CME Pune

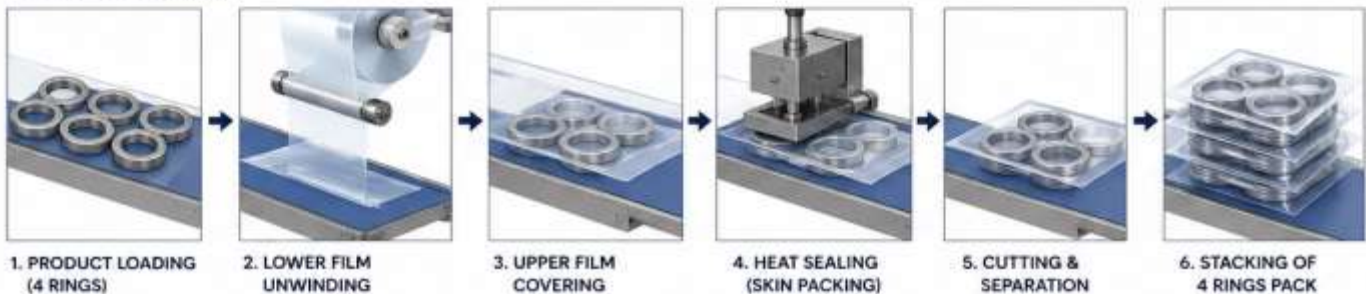
Piston Rings Packing Solution



HORIZONTAL SKIN PACKING MACHINE WITH STACKING OF THE 4 RINGS ON CONVEYOR



WORKING PROCESS



TECHNICAL SPECIFICATIONS

Machine Type	Horizontal Skin Packing
Product	4 Rings
Film Material	Transparent Skin Film
Sealing Type	Heat Sealing
Conveyor Type	Flat Belt Conveyor
Stacking	Automatic Stacking
Power Supply	3 Phase, 415V, 50Hz
Machine Frame	SS 304
Control System	PLC with HMI
Capacity	10-30 packs/min (Depends on product)

Chocolate Powder Packaging Solution



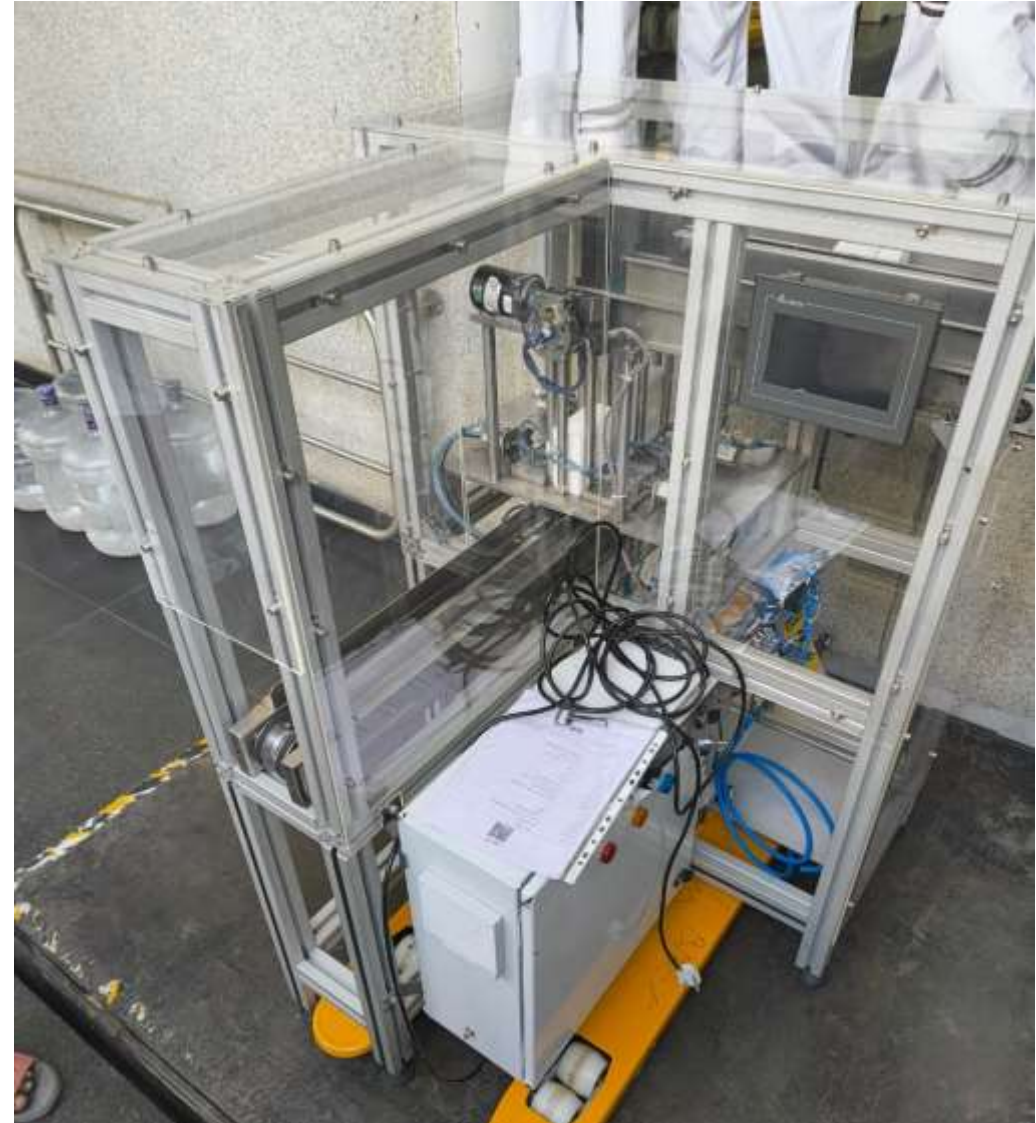
1. Servo Controlled Auger Mechanism
2. 250gm-1000gm packs variants
3. Min speed 60-120 ppm.
4. PLC-based control system.
5. Fabrication is in SS316
6. Fully washable.
7. 12" HMI for user convenience.
8. Power supply 440VAC



Vials Filling & Packaging Solution



1. Servo-controlled conveyor
2. Two-channel tray dispenser
3. Speed 400-800 vials/min
4. PLC-based control system.
5. Fabrication is in SS316
6. Fully washable.
7. 12" HMI for user convenience.
8. Power supply 440VAC



Makhana Packaging Solution



Potato Chips Packaging Solution





DEFENCE DRONES

Core Performance & Capabilities

A high-level overview of the platform's key metrics.

Key Performance Metrics



40-45 Min Endurance

Extended flight time for persistent ISR and long-range delivery missions.



15kg Overall Payload

High lift capacity for advanced sensors and multi-package delivery.



5 km Range

Long-range LOS operational control and reliable, high-quality video link.



Design: Airframe & Propulsion

- > **Airframe:** Compact, foldable design with a 200cm wheelbase for maximum stability and transportability.
- > **Propulsion:** Four 15kg-thrust rated motors provide extreme power and operational redundancy.
- > **Power:** Dual 6S 22,000 mAh LiPo pack configuration for reliable, high-discharge power.



Advanced ISR & Datalink



Real-Time Situational Awareness

The Aetherion H-15 is built for persistent overwatch, providing clear, actionable intelligence.

- * Up to 5km live video feed (variable 360p-1080p).
- * High-bandwidth 4K streaming at ~1km LOS.
- * Video quality dynamically adapts to link strength.



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The Offboard Compute Advantage

Redefining mission flexibility and onboard
efficiency.



The GCS: Your Control Center

Offboard Computational Power

All heavy computation (e.g., target tracking, dynamic replanning, motion detection) is performed at the Ground Control Station (GCS), not on the drone.

Onboard Simplicity

The drone requires no onboard computer. This radically reduces airborne weight, lowers power draw, and dramatically increases reliability by minimizing points of failure.



GCS: Full Mission Control

-  **Live Video:** Real-time monitoring and control feed.
-  **Mission Planning:** Autonomous waypoint upload and management.
-  **Manual Override:** Full joystick control for tactical maneuvers.
-  **Target Tracking:** Real-time automated tracking of selected targets.
-  **Dynamic Replan:** Update mission parameters in real-time from the GCS.

Autonomous & Advanced Ops

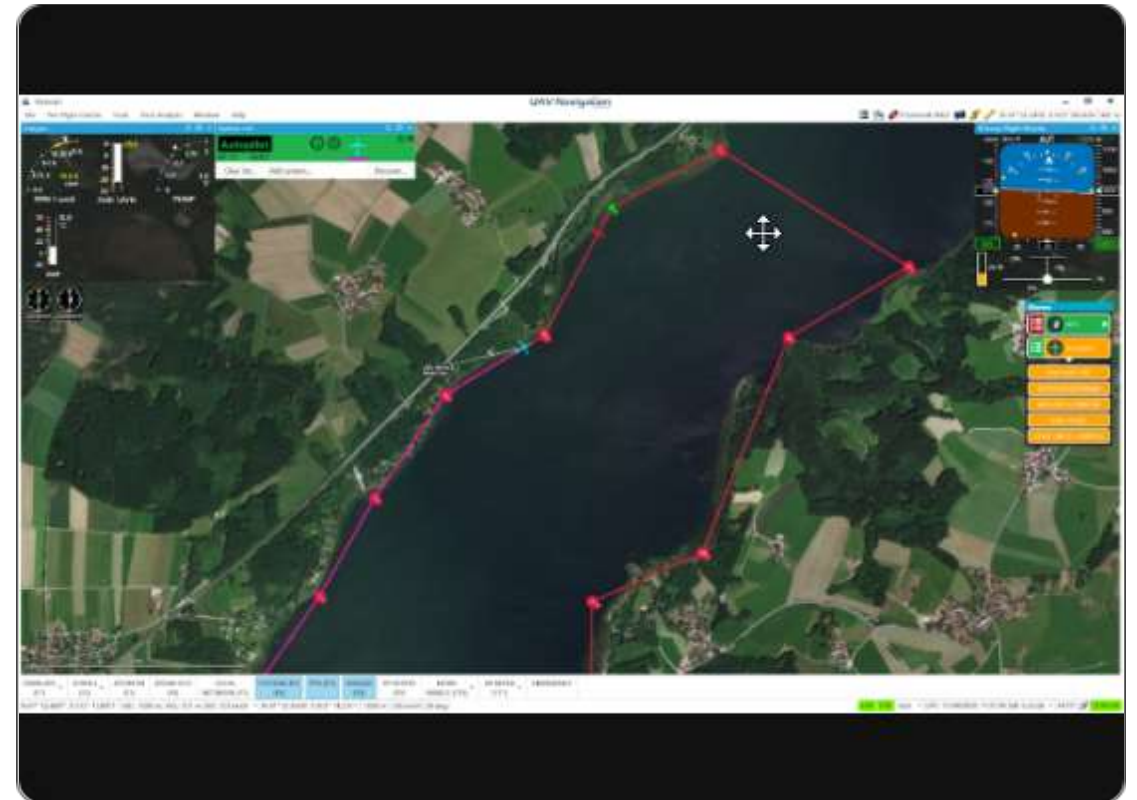


Flight Modes

- Loiter, Stabilize, Guided, plus fully autonomous VTOL takeoff, landing, and waypoint navigation.

Advanced Capabilities

- Automated motion detection.
- Persistent target tracking via GCS.
- Dynamic flight plan adjustment (retasking).



Reliability: Failsafe Systems



Comms Failure

Automatically executes a Return-to-Home (RTH) protocol upon loss of communication.



Low Battery

Triggers a safe and immediate auto-land protocol at critical battery levels.



Safe Altitudes

All failsafe behaviors are executed at user-configurable safe altitudes to avoid obstacles.

The Multi-Mission Advantage



Payload & Portability

High payload capacity (8kg) on a compact, transportable wheelbase.



Persistent ISR

Long endurance (45 mins) combined with a high-quality, long-range video feed.



Task Agnostic

Offboard compute makes the drone a flexible tool, adaptable for any mission.

Drone with Payload : Sensors & Instruments





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